

This document contains the derivation for the expression that is used to compute the chord function of an operator written in the position basis.

Let $|q_n\rangle$ and $|p_n\rangle$ be the position and momentum eigenstates on the torus, with Floquet angles $\chi_q = -2\pi\bar{p}$ and $\chi_p = -2\pi\bar{q}$:

$$|q_n + 1\rangle = e^{-i2\pi\bar{p}} |q_n\rangle, \quad |p_n + 1\rangle = e^{i2\pi\bar{q}} |p_n\rangle. \quad (1)$$

Consider an operator $\hat{\rho}$ written in the position representation:

$$\hat{\rho} = \sum_{nm} \rho_{nm} |q_n\rangle\langle q_m|, \quad (2)$$

with $0 \leq n, m \leq N - 1$, where N is the dimension of the Hilbert space on the torus, such that $2\pi\hbar N = 1$. Its chord function is obtained by expanding it on the basis of Weyl translation operators [1, 2]

$$\hat{\rho} = \frac{1}{\sqrt{N}} \sum_{\alpha} \mathcal{C}(\alpha) \hat{T}_{\alpha}, \quad (3)$$

where $\mathcal{C}(\alpha) = \text{Tr}(\hat{T}_{\alpha}^{\dagger} \hat{\rho}) / \sqrt{N}$ is the chord function which has support on the discrete lattice $\alpha = (k/N, l/N)$ with $k, l = 0, \dots, N - 1$.

From these expressions, when computing the chord function from the position representation [Eq. (2)] one gets

$$\mathcal{C}_{kl} \equiv \mathcal{C}(\alpha) = \frac{1}{\sqrt{N}} \sum_{nm} \rho_{nm} e^{i\frac{\pi}{N}kl} \langle q_m | \hat{T}_{\alpha}^{\dagger} | q_n \rangle \quad (4)$$

Using that $\hat{T}_z^{\dagger} = \hat{T}_{-z}^{\dagger}$ and $\hat{T}_{z+z'} = e^{\frac{i}{2\hbar}\sigma(z, z')} \hat{T}_z \hat{T}_{z'}$, where $\sigma(z, z') = z_1 z'_2 - z_2 z'_1$ is the symplectic product (here $z = (z_1, z_2)$), one can calculate

$$\langle q_m | \hat{T}_{\alpha}^{\dagger} | q_n \rangle = e^{i\frac{\pi}{N}kl} e^{-i2\pi l q_n} \langle q_m | q_{n-k} \rangle,$$

where $q_n = (n + \bar{q})/N$ and

$$\langle q_m | q_{n-k} \rangle = \begin{cases} \delta_{m, n-k} & \text{if } n - k \geq 0 \\ e^{i2\pi\bar{p}} \delta_{m, n-k-N} & \text{if } n - k < 0 \end{cases}.$$

Using this in Eq. (4) one gets

$$\boxed{\mathcal{C}_{kl} = \frac{1}{\sqrt{N}} e^{i\frac{\pi}{N}kl} e^{-i\frac{2\pi}{N}l\bar{q}} (A_{kl} + e^{i2\pi\bar{p}} B_{kl})} \quad (5)$$

where

$$A_{kl} = \sum_{n=k}^{N-1} e^{-i\frac{2\pi}{N}nl} \rho_{n, n-k} = e^{-i\frac{2\pi}{N}kl} \sum_{\nu=0}^{N-k-1} e^{-i\frac{2\pi}{N}\nu l} \rho_{\nu+k, \nu} \quad \text{and} \quad (6)$$

$$B_{kl} = \sum_{n=0}^{k-1} e^{-i\frac{2\pi}{N}nl} \rho_{n, n-k+N}, \quad (7)$$

which can be computed with FFTs.

References

- [1] Alfredo M. Ozorio de Almeida. “The Weyl representation in classical and quantum mechanics”. In: *Physics Reports* 295.6 (1998), pp. 265–342. ISSN: 0370-1573. DOI: [https://doi.org/10.1016/S0370-1573\(97\)00070-7](https://doi.org/10.1016/S0370-1573(97)00070-7). URL: <https://www.sciencedirect.com/science/article/pii/S0370157397000707>.
- [2] A.M. Ozorio de Almeida and O. Brodier. “Phase space propagators for quantum operators”. In: *Annals of Physics* 321.8 (2006), pp. 1790–1813. ISSN: 0003-4916. DOI: <https://doi.org/10.1016/j.aop.2006.03.007>. URL: <https://www.sciencedirect.com/science/article/pii/S0003491606000704>.